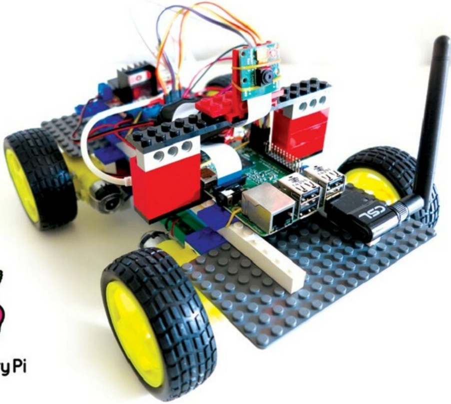
Top IDEs for Raspberry Pi

The Raspberry Pi, a tiny single-board computer, has revolutionised the way in which computer science is being taught in schools. It has also turned out to be a boon for software developers. Currently, it has gained popularity much beyond its target market and is being used in robotics projects.



Raspberry Pi, a small development board minicomputer that runs the Linux operating system, was developed in the United Kingdom by the Raspberry Pi Foundation to promote the teaching of basic computer science in schools in the UK and in developing countries. Raspberry Pi has USB sockets, which support various peripheral plug-and-play devices like the keyboard, the mouse, the printer, etc. It contains ports like HDMI (High Definition Multimedia Interface) to provide users with video output. Its credit-card-like size makes it extremely portable and affordable. It requires just a 5V micro-USB power supply, similar to the one used to charge a mobile phone.

Over the years, the Raspberry Pi Foundation has released a few different versions of the Pi board. The first version was Raspberry Pi 1 Model B, which was followed by a simple and cheap Model A. In 2014, the Foundation released a significant and improved version of the board —Raspberry Pi 1 Model B+. In 2015, the Foundation revolutionised the design of the board by releasing a small form factor edition costing US$ 5 (about ₹ 323) called Raspberry Pi Zero. In February 2016, Raspberry Pi 3 Model B was launched, which is currently the main product available. In 2017, the Foundation released the updated model of Raspberry Pi Zero named Raspberry Pi Zero W (W = wireless).

In the near future, a model that has improved technical specifications will arrive, offering a robust platform for embedded systems enthusiasts, researchers, hobbyists and engineers to use it in multi-functional ways to develop real-time applications.

Raspberry Pi as an efficient programming device

After getting the Pi powered up and the LXDE WM up and running, the user gets a full-fledged Linux box running a Debian based operating system, i.e., Raspbian. The Raspbian operating system comes with tons of free and open source utilities for users, covering programming, gaming, applications and even education.

The official programming language of Raspberry Pi is Python, which comes preloaded with the Raspbian operating system. The combination of Raspberry Pi and IDLE3, a Python integrated development environment, enables